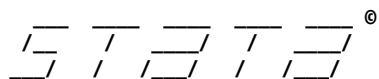


User: Analyse de ROA
Project: Analyse de ROA



17.0
MP-Parallel Edition

Statistics and Data Science

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Notes:

1. Unicode is supported; see [help unicode advice](#).
2. More than 2 billion observations are allowed; see [help obs advice](#).
3. Maximum number of variables is set to 5,000; see [help set maxvar](#).

1 . import excel "C:\Users\LENOVO\Downloads\Base De Donnée Final PFE.xlsx", sheet("ROA") firstrow
(9 vars, 50 obs)

2 . xtset BANQUES ANNEE, yearly

Panel variable: **BANQUES** (strongly balanced)
Time variable: **ANNEE, 2013 to 2022**
Delta: 1 year

3 .

4 . summarize ROA CNE RCNP ADC TB TC TMIC

Variable	Obs	Mean	Std. dev.	Min	Max
ROA	50	1.987813	6.994228	.0499049	49.87079
CNE	50	-8.924181	47.98103	-56.03281	61.92557
RCNP	50	3.121479	1.833508	1.479766	8.635694
ADC	50	23.89963	87.20499	5.494123	623.21
TB	50	8.221156	.5097876	6.080296	8.799629
TC	50	4.340557	6.24519	-7.18708	19.04728
TMIC	50	13.841	.4569966	13.09	14.39

5 .

6 . corr ROA CNE RCNP ADC TB TC TMIC
(obs=50)

	ROA	CNE	RCNP	ADC	TB	TC	TMIC
ROA	1.0000						
CNE	0.1477	1.0000					
RCNP	-0.0696	0.5109	1.0000				
ADC	0.9986	0.1609	-0.0642	1.0000			
TB	-0.6186	-0.5899	-0.5041	-0.6310	1.0000		
TC	-0.0199	0.1521	0.0504	-0.0345	-0.0306	1.0000	
TMIC	-0.0756	0.3042	0.1154	-0.0875	-0.0768	0.3570	1.0000

7 .

8 . xtsum ROA CNE RCNP ADC TB TC TMIC

Variable		Mean	Std. dev.	Min	Max	Observations	
ROA	overall	1.987813	6.994228	.0499049	49.87079	N =	50
	between		2.000421	.7711455	5.521837	n =	5
	within		6.756666	-3.316671	46.33677	T =	10
CNE	overall	-8.924181	47.98103	-56.03281	61.92557	N =	50
	between		41.39051	-51.70419	48.66334	n =	5
	within		30.0611	-76.43512	41.65595	T =	10
RCNP	overall	3.121479	1.833508	1.479766	8.635694	N =	50
	between		1.65892	1.940387	5.980411	n =	5
	within		1.056034	.3149551	5.776762	T =	10
ADC	overall	23.89963	87.20499	5.494123	623.21	N =	50
	between		27.70906	8.745375	73.21671	n =	5
	within		83.53408	-39.03041	573.893	T =	10
TB	overall	8.221156	.5097876	6.080296	8.799629	N =	50
	between		.4672898	7.649457	8.685633	n =	5
	within		.2857106	6.651995	8.484837	T =	10
TC	overall	4.340557	6.24519	-7.18708	19.04728	N =	50
	between		0	4.340557	4.340557	n =	5
	within		6.24519	-7.18708	19.04728	T =	10
TMIC	overall	13.841	.4569966	13.09	14.39	N =	50
	between		0	13.841	13.841	n =	5
	within		.4569966	13.09	14.39	T =	10

```

9 .
10 . *Calcul de SCR1 non contraint: Estimation équation par équation

11 .
12 . set more off

13 .
14 . local SCR1=0

15 .
16 . scalar N=5

17 .
18 . scalar TAN=10

19 .
20 . scalar K=6

21 .
22 . forvalues i=1/5 {
23 .   qui reg ROA CNE RCNP ADC TB TC TMIC if BANQUES==`i'
24 .   local SCR1=`SCR1'+e(rss)
25 . }

26 .
27 . di `SCR1'
    .9413593

28 .

```

```

29 . * Calcul de SCR1C contraint: Estimation sur le modèle empilé
30 .
31 . qui reg ROA CNE RCNP ADC TB TC TMIC
32 .
33 .
34 .
35 . local SCR1C=e(rss)
36 .
37 . di `SCR1C'
    4.7286475
38 .
39 . *Calcul de la statistique de Fisher F1 N=5 TAN=10 K=6
40 .
41 . local F1=((`SCR1C'-`SCR1')*(N*TAN-N*(K+1)))/(`SCR1'*(N-1)*(K+1))
42 .
43 . di `F1'
    2.1552922
44 .
45 . *La P_value de F1
46 .
47 . di "dof1 = " (N-1)*(K+1) " dof2 = " (N*TAN-N*(K+1))
    dof1 = 28 dof2 = 15
48 .
49 . local PVF1=Ftail((K+1)*(N-1),(N*TAN-N*(K+1)),`F1')
50 .
51 . di invFtail((K+1)*(N-1),(N*TAN-N*(K+1)),0.05)
    2.2586576
52 .
53 .
54 .
55 . * Calcul de SCR1CP: estimation du modèle à effets individuels
56 .
57 . xtreg ROA CNE RCNP ADC TB TC TMIC,fe

```

```

Fixed-effects (within) regression      Number of obs   =      50
Group variable:  BANQUES              Number of groups =       5

R-squared:                            Obs per group:
    Within = 0.9992                    min =      10
    Between = 0.9707                   avg  =     10.0
    Overall = 0.9929                   max  =      10

                                F(6,39)      =     7686.26
corr(u_i, Xb) = -0.2679              Prob > F      =     0.0000

```

ROA	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
CNE	.0002077	.0015825	0.13	0.896	-.0029933	.0034086
RCNP	-.0416607	.0395131	-1.05	0.298	-.1215834	.038262
ADC	.0775952	.0009537	81.36	0.000	.0756662	.0795242
TB	-1.063704	.2830572	-3.76	0.001	-1.636241	-.4911669
TC	.0128679	.0054169	2.38	0.023	.0019112	.0238247
TMIC	-.0023902	.0876121	-0.03	0.978	-.1796024	.174822
_cons	8.987317	2.973834	3.02	0.004	2.972171	15.00246
sigma_u	.63590899					
sigma_e	.2201475					
rho	.89297683	(fraction of variance due to u_i)				

F test that all u_i=0: F(4, 39) = 14.64

Prob > F = 0.0000

```

58 .
59 . local SCR1CP=e(rss)

60 .
61 . di `SCR1CP'
    1.890132

62 .
63 . *Calcul de la statistique de Fisher F2

64 .
65 . local F2=((`SCR1CP'-`SCR1')*(N*TAN-N*(K+1)))/(`SCR1'*(N-1)*K)

66 .
67 .
68 .
69 . *La P_value de F2

70 .
71 . di "dof1 = " K*(N-1) " dof2 = " (N*TAN-N*(K+1))
    dof1 = 24 dof2 = 15

72 .
73 . local PVF2=Ftail(K*(N-1),(N*TAN-N*(K+1)),`F2')

74 .
75 .
76 .
77 . *Calcul de la statistique de Fisher F3

78 .
79 . local F3=(`SCR1C'-`SCR1CP')*(N*(TAN-1)-K)/(`SCR1CP'*(N-1))

80 .
81 .
82 .
83 . *La P_value de F3

84 .
85 . di "dof1 = " (N-1) " dof2 = " (N*(TAN-1)-K)
    dof1 = 4 dof2 = 39

86 .
87 . local PVF3=Ftail((N-1),(N*(TAN-1)-K),`F3')

88 .
89 . *Affichage des résultats

90 .
91 . di in y " SCR1 = " in gr `SCR1'
    SCR1 = .9413593

92 .
93 . di in y " SCR1C = " in gr `SCR1C'
    SCR1C = 4.7286475

94 .
95 . di in y " SCR1CP = " in gr `SCR1CP'
    SCR1CP = 1.890132

96 .
97 .

```

```

98 .
99 . di in y "F1 = " in gr `F1'
    F1 = 2.1552922

100 .
101 . di in y "F2 = " in gr `F2'
    F2 = .62992201

102 .
103 . di in y "F3 = " in gr `F3'
    F3 = 14.642113

104 .
105 .
106 .
107 . di in y "PvalF1 = " in gr `PVF1'
    PvalF1 = .06020823

108 .
109 . di in y "PvalF2 = " in gr `PVF2'
    PvalF2 = .84833713

110 .
111 . di in y "PvalF3 = " in gr `PVF3'
    PvalF3 = 2.179e-07

112 .
113 .
114 .
115 . xtreg ROA CNE RCNP ADC TB TC TMIC, fe

```

```

Fixed-effects (within) regression      Number of obs   =      50
Group variable:  BANQUES              Number of groups =       5

R-squared:                             Obs per group:
    Within = 0.9992                    min =      10
    Between = 0.9707                   avg  =     10.0
    Overall = 0.9929                   max  =      10

corr(u_i, Xb) = -0.2679                F(6,39)          =    7686.26
                                         Prob > F          =    0.0000

```

ROA	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
CNE	.0002077	.0015825	0.13	0.896	-.0029933	.0034086
RCNP	-.0416607	.0395131	-1.05	0.298	-.1215834	.038262
ADC	.0775952	.0009537	81.36	0.000	.0756662	.0795242
TB	-1.063704	.2830572	-3.76	0.001	-1.636241	-.4911669
TC	.0128679	.0054169	2.38	0.023	.0019112	.0238247
TMIC	-.0023902	.0876121	-0.03	0.978	-.1796024	.174822
_cons	8.987317	2.973834	3.02	0.004	2.972171	15.00246
sigma_u	.63590899					
sigma_e	.2201475					
rho	.89297683	(fraction of variance due to u_i)				

```

F test that all u_i=0: F(4, 39) = 14.64                Prob > F = 0.0000

```

```

116 .
117 . estimate store fe

```

118 .
 119 . xtreg ROA CNE RCNP ADC TB TC TMIC, re

Random-effects GLS regression Number of obs = 50
 Group variable: **BANQUES** Number of groups = 5

R-squared: Obs per group: min = 10
 Within = **0.9985** avg = 10.0
 Between = **0.9943** max = 10
 Overall = **0.9980**

corr(u_i, X) = 0 (assumed) Wald chi2(6) = 21754.52
 Prob > chi2 = 0.0000

ROA	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
CNE	-.0022666	.0013737	-1.65	0.099	-.004959	.0004258
RCNP	.0475605	.0368914	1.29	0.197	-.0247453	.1198663
ADC	.0816051	.0008757	93.19	0.000	.0798889	.0833214
TB	.3014341	.1877599	1.61	0.108	-.0665685	.6694367
TC	.0141475	.0081352	1.74	0.082	-.0017973	.0300922
TMIC	.212988	.1165589	1.83	0.068	-.0154632	.4414392
_cons	-5.618718	2.288376	-2.46	0.014	-10.10385	-1.133582
sigma_u	0					
sigma_e	.2201475					
rho	0	(fraction of variance due to u_i)				

120 .
 121 . estimate store re
 122 .
 123 . hausman fe re

	Coefficients		(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
	(b) fe	(B) re		
CNE	.0002077	-.0022666	.0024743	.0007857
RCNP	-.0416607	.0475605	-.0892212	.014153
ADC	.0775952	.0816051	-.0040099	.0003778
TB	-1.063704	.3014341	-1.365138	.2118197
TC	.0128679	.0141475	-.0012796	.
TMIC	-.0023902	.212988	-.2153782	.

b = Consistent under H0 and Ha; obtained from **xtreg**.
 B = Inconsistent under Ha, efficient under H0; obtained from **xtreg**.

Test of H0: Difference in coefficients not systematic

chi2(6) = (b-B)'[(V_b-V_B)^(-1)](b-B)
 = 42.52
 Prob > chi2 = 0.0000
 (V_b-V_B is not positive definite)

124 .
 125 . xttest0

Breusch and Pagan Lagrangian multiplier test for random effects

ROA[BANQUES,t] = Xb + u[BANQUES] + e[BANQUES,t]

Estimated results:

	Var	SD = sqrt(Var)
ROA	48.91922	6.994228
e	.0484649	.2201475
u	0	0

Test: Var(u) = 0

chibar2(01) = 0.00
Prob > chibar2 = 1.0000

126 .

127 . xtreg ROA CNE RCNP ADC TB TC TMIC, fe

Fixed-effects (within) regression Number of obs = 50
Group variable: BANQUES Number of groups = 5

R-squared: Obs per group:
 Within = 0.9992 min = 10
 Between = 0.9707 avg = 10.0
 Overall = 0.9929 max = 10

corr(u_i, Xb) = -0.2679 F(6,39) = 7686.26
 Prob > F = 0.0000

ROA	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
CNE	.0002077	.0015825	0.13	0.896	-.0029933	.0034086
RCNP	-.0416607	.0395131	-1.05	0.298	-.1215834	.038262
ADC	.0775952	.0009537	81.36	0.000	.0756662	.0795242
TB	-1.063704	.2830572	-3.76	0.001	-1.636241	-.4911669
TC	.0128679	.0054169	2.38	0.023	.0019112	.0238247
TMIC	-.0023902	.0876121	-0.03	0.978	-.1796024	.174822
_cons	8.987317	2.973834	3.02	0.004	2.972171	15.00246
sigma_u	.63590899					
sigma_e	.2201475					
rho	.89297683	(fraction of variance due to u_i)				

F test that all u_i=0: F(4, 39) = 14.64 Prob > F = 0.0000

128 .

129 . predict residu, resid

130 .

131 . xttest3

Modified Wald test for groupwise heteroskedasticity
in fixed effect regression model

H0: sigma(i)^2 = sigma^2 for all i

chi2 (5) = 8.80
Prob>chi2 = 0.1175

132 .

133 . sktest residu

Skewness and kurtosis tests for normality

Variable	Obs	Pr(skewness)	Pr(kurtosis)	Joint test	
				Adj chi2(2)	Prob>chi2
residu	50	0.1812	0.0530	5.34	0.0692

134 .

135 . xtserial residu

Wooldridge test for autocorrelation in panel data

H0: no first-order autocorrelation

F(1, 4) = 8.954
Prob > F = 0.0402

```
136 .
137 . tabulate BANQUES, generate(z)
```

BANQUES	Freq.	Percent	Cum.
1	10	20.00	20.00
2	10	20.00	40.00
3	10	20.00	60.00
4	10	20.00	80.00
5	10	20.00	100.00
Total	50	100.00	

```
138 .
139 . xtglm ROA CNE RCNP ADC TB TC TMIC z1 z2 z3 z4 z5 ,noconstant corr(ar1)
```

Cross-sectional time-series FGLS regression

Coefficients: **generalized least squares**

Panels: **homoskedastic**

Correlation: **common AR(1) coefficient for all panels (0.1487)**

Estimated covariances	=	1	Number of obs	=	50
Estimated autocorrelations	=	1	Number of groups	=	5
Estimated coefficients	=	11	Time periods	=	10
			Wald chi2(11)	=	72472.26
			Prob > chi2	=	0.0000

ROA	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
CNE	.0002711	.0014959	0.18	0.856	-.0026608	.003203
RCNP	-.0403581	.0370653	-1.09	0.276	-.1130048	.0322885
ADC	.0773757	.0008061	95.99	0.000	.0757959	.0789555
TB	-1.121344	.2411755	-4.65	0.000	-1.59404	-.6486491
TC	.0122316	.0044694	2.74	0.006	.0034717	.0209915
TMIC	-.0275935	.0818181	-0.34	0.736	-.187954	.132767
z1	10.52032	2.670224	3.94	0.000	5.286775	15.75386
z2	9.419854	2.505939	3.76	0.000	4.508305	14.3314
z3	10.08203	2.652807	3.80	0.000	4.882623	15.28143
z4	8.877312	2.491298	3.56	0.000	3.994457	13.76017
z5	10.18487	2.615459	3.89	0.000	5.058667	15.31108

```
140 .
141 . xtglshr
```

Cross-sectional time-series FGLS regression

Coefficients: **generalized least squares**

Panels: **homoskedastic**

Correlation: **common AR(1) coefficient for all panels (0.1487)**

Estimated covariances	=	1	Number of obs	=	50
Estimated autocorrelations	=	1	Number of groups	=	5
Estimated coefficients	=	11	Time periods	=	10
			Wald chi2(11)	=	72472.26
			Prob > chi2	=	0.0000

(Std. err. adjusted for 50 clusters in Individual_Observations)

ROA	Coefficient	Robust std. err.	z	P> z	[95% conf. interval]	
CNE	.0002711	.0015327	0.18	0.860	-.002733	.0032751
RCNP	-.0403581	.0358737	-1.13	0.261	-.1106692	.0299529
ADC	.0773757	.0003595	215.26	0.000	.0766712	.0780802
TB	-1.121344	.1019767	-11.00	0.000	-1.321215	-.9214737
TC	.0122316	.0044668	2.74	0.006	.0034768	.0209864
TMIC	-.0275935	.0832785	-0.33	0.740	-.1908163	.1356293
z1	10.52032	1.615699	6.51	0.000	7.353606	13.68703
z2	9.419854	1.596027	5.90	0.000	6.2917	12.54801
z3	10.08203	1.652289	6.10	0.000	6.843602	13.32045
z4	8.877312	1.553122	5.72	0.000	5.833249	11.92137

z5	10.18487	1.616434	6.30	0.000	7.016721	13.35303
----	----------	----------	------	-------	----------	----------

142 .